

Home Work

Ex 1

$$\int_1^2 \frac{x+2}{x^2+4x-1} dx \Rightarrow \int f(g(x)) g'(x) dx$$
$$\Rightarrow \int f(t) dt$$

$$t = x^2 + 4x - 1$$

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$$\frac{dt}{dx} = 2x + 4 = 2(x+2) \Rightarrow dt = 2(x+2) dx$$

$$\Rightarrow (x+2)^{dx} = \frac{1}{2} dt = f(t) dt$$

$$\text{So When } x=1 \Rightarrow t = 1^2 + 4 - 1 = 4$$

$$\text{" } x=2 \Rightarrow t = (2)^2 + 4(2) - 1 = 11$$

$$\text{So } \int_1^2 \frac{x+2}{x^2+4x-1} dx \Rightarrow \int_4^{11} \frac{1}{2t} dt = \frac{1}{2} \int_4^{11} \frac{1}{t} dt$$

$$= \frac{1}{2} \left(\left[\ln(t) \right]_4^{11} - \left[\ln(t) \right]_4^{11} \right)$$

$$= \frac{1}{2} (\ln 11 - \ln 4)$$

$$= \frac{1}{2} \ln \frac{11}{4}$$

Ex 2

Using integration to find $F(x)$ of $f(x) = \ln(x+1)$
for $F(0) = 0$

$$F(x) = \int \ln(x+1)$$

$$= x \ln(x+1) - \int \frac{x+1-1}{x+1}$$

$$= x \ln(x+1) - \int 1 - \frac{1}{x+1}$$

$$= x \ln(x+1) + x + \ln(x+1)$$

$$= (x+1) \ln(x+1) + x$$

Ex 3.

$$1) \int \frac{dx}{x^2 - x - 2} = \int \frac{dx}{(x-2)(x+1)}$$

$$\frac{(x-2)(x+1)}{(x-2)(x+1)} = \left(\frac{a}{x-2} + \frac{b}{x+1} \right) (x+1)(x-2)$$

$$1 = a(x+1) + b(x-2)$$

$$1 = ax + a + bx - 2b$$

$$1 = (a+b)x + a - 2b$$

$$\text{So } a+b=0 \Rightarrow a=-b$$

$$a-2b=1 \Rightarrow -b-2b=1 \Rightarrow -3b=1 \Rightarrow b=-\frac{1}{3}$$

$$\text{So } a = \frac{1}{3}$$

$$\text{So } \frac{1}{(x-2)(x+1)} = \frac{1}{3(x-2)} - \frac{1}{3(x+1)}$$

$$\begin{aligned} \Rightarrow \int \frac{dx}{x^2 - x - 2} &= \frac{1}{3} \int \frac{1}{x-2} - \frac{1}{3} \int \frac{1}{x+1} \\ &= \frac{1}{3} \ln(x-2) - \frac{1}{3} \ln(x+1) \\ &= \frac{1}{3} \ln\left(\frac{x-2}{x+1}\right) \end{aligned}$$

$$2) \int \frac{2x+1}{x^2-2x+1} dx$$

$$= \int \frac{2x+1}{(x-1)^2} dx$$

$$t = (x-1)^2 \Rightarrow x = t+1$$

$$dx = 1 dt$$

So

$$= \int \frac{2(t+1)+1}{t^2} dt = \int \frac{2t+3}{t^2} dt = \int \frac{2}{t} + \int \frac{3}{t^2} dt$$

$$= 2 \int \frac{1}{t} + 3 \int \frac{1}{t^2} = 2 \ln t + \left(-\frac{3}{t} \right)$$

$$= 2 \ln(x-1) - \frac{3}{x-1} + C$$

$$3) \int \frac{2x+3}{x^2+x+1} dx = \ln(x^2+x+1) + \int \frac{2}{x^2+x+1} dx$$

$$= \ln(x^2+x+1) + \int \frac{2}{\left(x+\frac{1}{2}\right)^2 + \frac{3}{4}} dx = \ln(x^2+x+1) + \int \frac{2}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx$$

$$\text{We put } x + \frac{1}{2} = \frac{\sqrt{3}}{2} \tan \theta$$

$$x = \frac{\sqrt{3}}{2} \tan \theta - \frac{1}{2} \Rightarrow dx = \frac{\sqrt{3}}{2} \frac{1}{\cos^2 \theta} d\theta = \frac{\sqrt{3}}{2} \sec^2 \theta d\theta$$

So

$$= \ln(x^2+x+1) + \int \frac{2}{\left(\frac{\sqrt{3}}{2} \tan \theta\right)^2 + \frac{3}{4}} \cdot \frac{\sqrt{3}}{2} \sec^2 \theta d\theta$$

$$= \ln(x^2+x+1) + \int \frac{2}{\frac{3}{4} \tan^2 \theta + \frac{3}{4}} \cdot \frac{\sqrt{3}}{2} \sec^2 \theta d\theta$$

$$= \ln(x^2+x+1) + \int \frac{2}{\frac{3}{4} (\tan^2 \theta + 1)} \cdot \frac{\sqrt{3}}{2} \sec^2 \theta d\theta = \ln(x^2+x+1) + \int \frac{4\sqrt{3} \sec^2 \theta}{3 (\sec^2 \theta)} d\theta$$

$$= \ln(x^2+x+1) + \frac{4\sqrt{3}}{3} \theta = \ln(x^2+x+1) + \frac{4\sqrt{3}}{3} \arctan\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

Ex 4.

$$2) A = \int \sqrt{x^2 + 2} \, dx$$

$$= x \sqrt{x^2 + 2} - \int x \frac{2x}{2\sqrt{x^2 + 2}} \, dx$$

$$= x \sqrt{x^2 + 2} - \int \frac{x^2 + 2 - 2}{\sqrt{x^2 + 2}} \, dx$$

$$= x \sqrt{x^2 + 2} + \int \sqrt{x^2 + 2} \, dx + \int \frac{2}{\sqrt{x^2 + 2}} \, dx =$$

$$A = x \sqrt{x^2 + 2} - A + \int \frac{2}{\sqrt{x^2 + 2}} \, dx \Rightarrow 2A = x \sqrt{x^2 + 2} + \int \frac{2}{\sqrt{x^2 + 2}} \, dx$$

$$2A = x \sqrt{x^2 + 2} + 2 \int \frac{1}{\sqrt{x^2 + 2}} \, dx \Rightarrow A = \frac{x}{2} \sqrt{x^2 + 2} + \ln(x + \sqrt{x^2 + 2})$$

$$2) B = \int x \sqrt{-x^2 + x + 1} dx$$

$$= \int x \sqrt{-(x^2 - x - 1)} dx = \int x \sqrt{\left(x - \frac{1}{2}\right)^2 - \frac{5}{4}} dx$$

$$= \int x \sqrt{\frac{5}{4} - \left(x - \frac{1}{2}\right)^2} dx$$

Put $t = x - \frac{1}{2} \Rightarrow x = t + \frac{1}{2}$

$$B = \int \left(t + \frac{1}{2}\right) \sqrt{\frac{5}{4} - t^2} dt = \int t \left(\sqrt{\frac{5}{4} - t^2}\right) dt + \frac{1}{2} \int \sqrt{\frac{5}{4} - t^2} dt$$

$$= -\frac{1}{3} \left(\frac{5}{4} - t^2\right)^{3/2} + \frac{t}{2} \sqrt{\frac{5}{4} - t^2} + \frac{5}{8} \arcsin\left(\frac{2t}{\sqrt{5}}\right)$$

$$= -\frac{1}{3} (-x^2 + x + 1) + \frac{x - \frac{1}{2}}{4} \sqrt{-x^2 + x + 1} + \frac{5}{16} \arcsin\left(\frac{2x - 1}{\sqrt{5}}\right) + C$$

$$\begin{aligned}
 1) \int \sin 2x \sin 3x \, dx &= \frac{1}{2} \int [\cos(2x-3x) - \cos(2x+3x)] \\
 &= \frac{1}{2} \int [\cos(-x) - \cos(5x)] \, dx \\
 &= \frac{1}{2} \left(\int \cos(x) \, dx - \int \cos(5x) \, dx \right) \\
 &= \frac{1}{2} \left(\sin x - \frac{\sin 5x}{5} \right) \\
 &= \frac{1}{2} \sin x - \frac{1}{10} \sin 5x + C
 \end{aligned}$$

$$2) I = \int \sin^3 x \cos^2 x \, dx = \int \sin^2 x (1 - \cos^2 x) \cos^2 x \, dx$$

$$\begin{aligned}
 \text{Put } t = \cos x &\rightarrow dt = -\sin x \, dx \\
 &\rightarrow -dt = \sin x \, dx
 \end{aligned}$$

So

$$\begin{aligned}
 I &= \int (1-t^2) t^2 (-dt) = \int (t^4 - t^2) \, dt \\
 &= \frac{1}{5} t^5 - \frac{1}{3} t^3 \\
 &= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C
 \end{aligned}$$

$$\begin{aligned}
 3) C &= \int \frac{1}{5-4\cos x} \, dx = \int \frac{1}{5-4\left(\frac{1+t^2}{1+t^2}\right)} \cdot \frac{2}{1+t^2} \, dt \\
 &= \int \frac{1}{\frac{5(1+t^2)-4(1+t^2)}{1+t^2}} \cdot \frac{2}{1+t^2} \, dt \\
 &= 2 \int \frac{1}{1+9t^2} \, dt = \\
 &= \frac{2}{3} \arctan(3t) + C \\
 &= \frac{2}{3} \arctan\left(3 \tan \frac{x}{2}\right) + C
 \end{aligned}$$